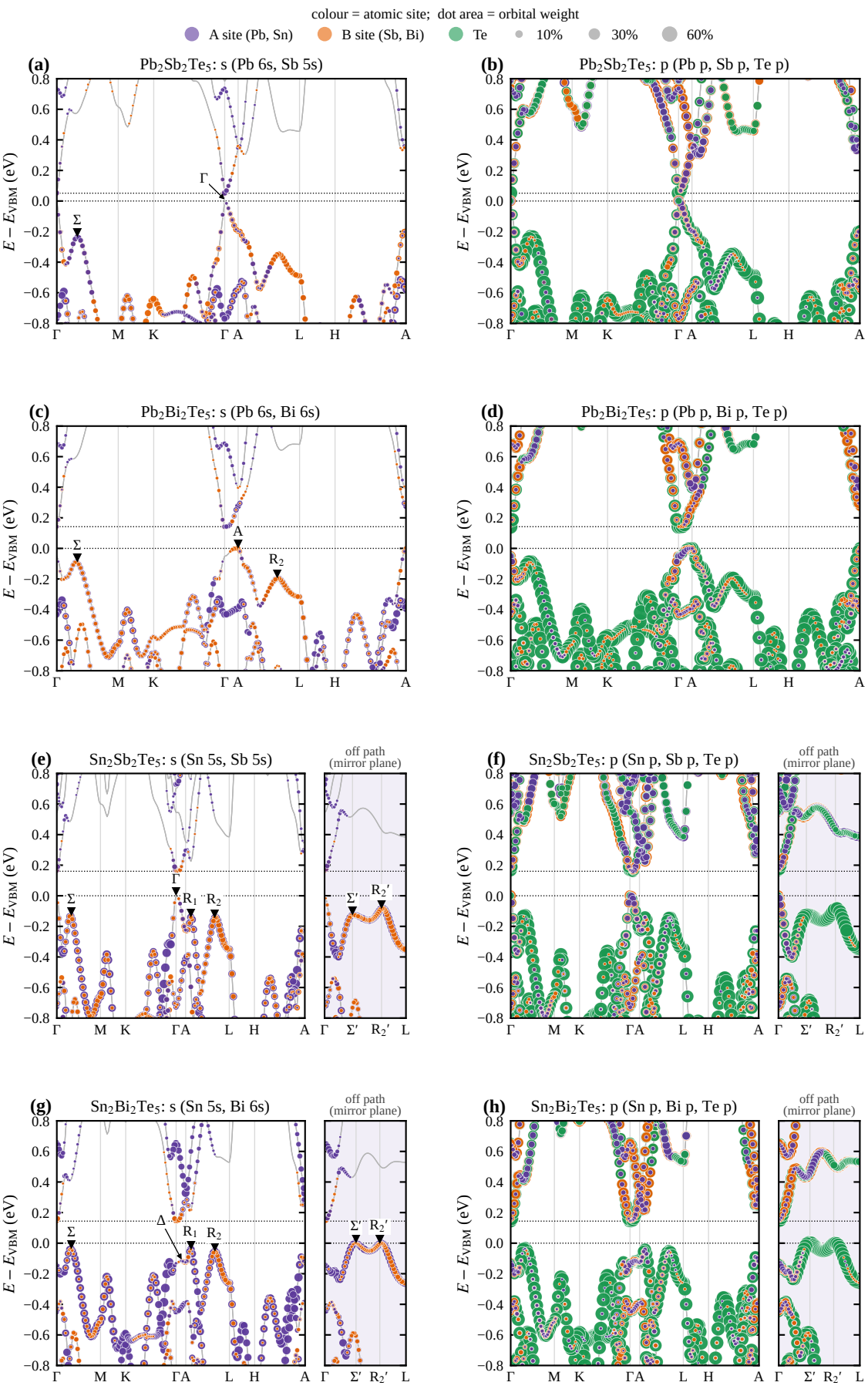

colour = atomic site; dot area = orbital weight
A site (Pb, Sn)
B site (Sb, Bi)
Te
10%
30%
60%
(a)
Pb2Sb2Te5: s (Pb 6s, Sb 5s)
(b)
Pb2Sb2Te5: p (Pb p, Sb p, Te p)
(c)
Pb2Bi2Te5: s (Pb 6s, Bi 6s)
(d)
Pb2Bi2Te5: p (Pb p, Bi p, Te p)
(e)
Sn2Sb2Te5: s (Sn 5s, Sb 5s)
(f)
Sn2Sb2Te5: p (Sn p, Sb p, Te p)
(g)
Sn2Bi2Te5: s (Sn 5s, Bi 6s)
(h)
Sn2Bi2Te5: p (Sn p, Bi p, Te p)
off path (mirror plane)
E − EVBM (eV)
Γ M K Γ A L H A
Γ M K ΓA L H A
Γ Σ′ R2′ L